

Expanding computing power and connectivity is allowing manufacturers to think big and develop more ambitious goals and objectives for their work. Further, advanced robotic processes are helping remanufacturers to successfully process a higher percentage of product components and materials.

Robots have already proven their capabilities in remanufacturing under certain conditions with relatively small and simple batches of components that usually involve some significant human inspection. Advanced artificial intelligence (AI) has resulted in new patterns in human-machine interactions.

Industries always aim to improve efficiency, increase productivity and reduce costs. The IoT connectivity and other elements of AI are helping improve predictive maintenance and avoid machine failure during critical phases of production.

One of the significant challenges in remanufacturing is inventory forecasting. Remanufacturers should be able to predict the demand for returned items. AI technologies can greatly improve existing forecasting models to predict product returns.

Elements of Big Data and machine learning can make real-time analysis of data on sales, product usage and warranty activity. These analyses help predict accurately product life expectancy, rate and timing of return into the remanufacturing process. Thus, automation in remanufacturing is making a good impact on economic and environmental stability.

The remanufacturing industry mainly faces problems of timely, accurate and consistent information of remanufacturing resources.

Recent automation technologies like AI, machine learning, the IoT, etc can be used to solve this problem. Under the IoT, an identification technology for disassembled product parts is designed, and real-time status of remanufacturable resources are monitored. Based on captured remanufacturing information, a real-time production scheduling method is developed. A model is then developed to achieve



Remanufactured drum unit (Credit: <https://is.alicdn.com>)

cost-reduction for dynamic management of manufacturable resources.

With remanufacturing, a machine part, perhaps from an aircraft engine or a mining truck engine, can be restored through an industrial process, potentially using cobots (collaborative robots). In the remanufacturing process, however, shape or geometry of the input part is different each time. When it comes out of service, we do not know how it will look. And with advancements in robotics, combined with latest technologies such as 3D-scanning systems, robots can be tuned to respond according to variations in input conditions.

This flexibility and adaptability would also mean that robots can now be put to work on smaller batches with high-mix low-volume manufacturing, opening up the option of automation for smaller manufacturers.

Advancements in Industry 4.0 have resulted in the immense potential to unlock the power of remanufacturing. **Smart sensors, cloud computing, robotics, machine-to-machine communication, additive manufacturing and others are some of the key contributors to the remanufacturing industry.**

Industry 4.0 has improved data transferability by building knowledge/data-sharing platforms through sensors, embedded systems and the IoT. This has greatly benefitted the remanufacturing industry, too. Even smart factories that provide high flexibility and small batch size production seamlessly address these complex matters associated with the remanufacturing process.

Today, Big Data mining, particle swarm optimisation and back propagation neural network algorithms are

being utilised to implement cost reduction in remanufacturing processes. Data mining technology can be used to optimise remanufacturing process management, including achieving dimensional accuracy to overcome uncertainties for reassembly dimensions and so on.

What India wants in remanufacturing

The remanufacturing industry is in its infancy in India. The country is mostly unaware of

the rapid steps that the remanufacturing industry has put forward across various countries of the world such as the US, the EU countries, Australia, China, Singapore and so on. Moreover, the government of India has not been greatly supportive towards this industry. Import of remanufactured goods is banned as it is considered to be of low quality.

“Given the present low awareness and poor penetration of remanufacturing across India, there is a huge opportunity for growth of remanufacturing. However, we have to improve awareness around remanufacturing and showcase the various benefits it yields to the overall economy and environment,” says Satish Rao, an electrical engineer and supplier-cum-dealer for Caterpillar.

Finally

Compared with manufacturing, remanufacturing uses as little as ten per cent energy and raw materials, while lowering carbon-dioxide emissions by more than eighty per cent.

Today, a broad range of firms, including Caterpillar and GE, are engaged in remanufacturing of products. Earth moving, automotive parts, electronics, medical devices and information technology are major industries in this sector that have made a mark globally.

We are living in the period of global climate change. The remanufacturing industry can reduce carbon-emissions and provide better environmental safety to countries and consumers. For large global consumers, remanufactured products are generally of high quality with similar life expectancy as new goods. **EFY**